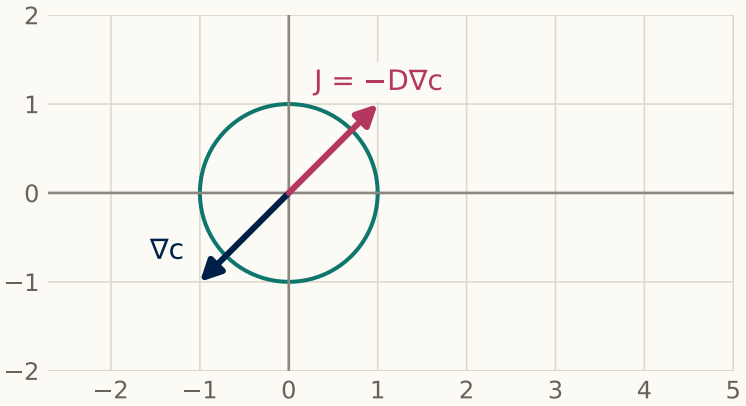
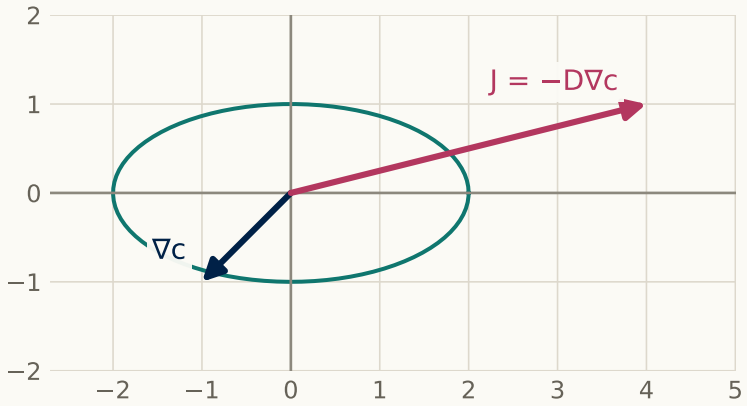
Diffusion can favour one direction

isotropic: D = I



anisotropic: D = diag(4,1)



At the same time, concentration contours have axis ratio $\sqrt{(D_x/D_y)}$.